

Forecasting Capability Emergence in Language Models

A Multi-Seed, Negative-Certified Benchmark at Scale

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1. Research problem to be solved

Frontier labs cannot predict *when* a training run will acquire a specific capability. Aggregate loss falls smoothly while individual abilities — in-context learning, arithmetic, multi-step reasoning — appear abruptly. This unpredictability is a recognized safety problem: the 2026 International AI Safety Report lists capability forecastability as open, and existing work does not close it. Prior approaches predict across *model scale* (PassUntil, emergence laws by finetuning), or offer early-warning *indicators* without forecasting discipline (spectral observables, loss-landscape signals, stagewise detection via the local learning coefficient), or predict a single formation time *per configuration* from pre-training hyperparameters (Aoyama & Wilcox 2025). None issues a scored, calibrated forecast for a given training run, and none reports false-alarm rates — because no public checkpoint suite contains the negative runs required to measure them.

We have closed this gap at small scale, on one consumer GPU, and the result is strong:

- **Per-seed forecasting works.** Across 30 transformers at *identical* configuration — where the config-equation baseline predicts one constant by construction — the formation time of a mechanistic precursor circuit (the previous-token head, the known antecedent of induction) forecasts each seed’s induction-head emergence at Spearman $\rho = 0.977$ (95% CI [0.911, 0.995]), with a median lead of 975 steps ($\approx 15\%$ of training).
- **Loss curves cannot do this.** A best-case training-loss rule, granted an oracle threshold sweep, *ties* that ranking ($\rho = 0.977$) with a 50-step median lead: it detects the cliff in progress. Warning time comes from *where you anchor*, and only the circuit precursor anchors early.
- **Calibrated, not heuristic.** Split-conformal intervals of width 300 steps, anchored at the precursor alarm, covered 15/15 held-out seeds at 90% nominal.
- **Blind-validated.** The entire rule was frozen verbatim and commit-stamped, then tested on a never-seen configuration (wider model, sparser data structure): 10/10 interval coverage (pre-registered bar 7/10), 10/10 pre-event alarms, 0/3 false alarms, $\rho = 0.988$.
- **False alarms certified.** Every false-alarm claim is measured against 21 *manufactured capability-blocked negatives* — runs constructed so the capability *cannot* form (training signal ablated; architecture incapable). Zero false alarms across all rungs.
- **Public suites cannot answer this.** We probed five Pythia suites (70M/160M/410M $\times$ standard/deduped): the precursor leads the induction cliff by a full checkpoint stage in every suite — and all five cliff at the same token count, confirming config-determined timing. With one run per configuration and no negatives, public artifacts structurally cannot test per-seed forecasting or false-alarm rates.

What this grant buys: the scale-up from synthetic languages and 2.7M-parameter models to real language-model pretraining (70M–1B parameters, real text corpora, multiple capability tracks),

and the public release of the first multi-seed emergence corpus with manufactured negatives. The methodology is proven; what is missing is compute. A single RTX 3080 cannot train 30-seed fleets at 1B parameters — our current fleet of 45 runs took 11 hours at 2.7M parameters; the same experiment at 410M is ~ 3 GPU-months on that hardware and ~ 2 days on a Nebius H100 node.

2. Expected resource usage

Workstream	GPU-h (H100)	Tokens	Notes
Multi-seed fleets, 70M–410M	1,500	$\sim 300\text{B}$	30 seeds $\times$ 3 configs, $\sim 10\text{B}$ tok/run
Manufactured negatives (same scales)	700	$\sim 140\text{B}$	data-ablated + architectural, $\geq 10/\text{class}$
Capability tracks 2–3 (ICL, arithmetic)	2,500	$\sim 400\text{B}$	incl. precursor-identification probing
1B validation fleet	1,200	$\sim 100\text{B}$	10 seeds + 3 negatives, blind gate
Gap-origin scaling study	800	$\sim 150\text{B}$	$\text{lr} \times \text{batch} \times \text{width}$ factorial
Design-iteration headroom ($\sim 1.5\times$)	3,000	—	empirical rate from our program
Total	$\sim 9,700$	$\sim 1.1\text{T}$	

Storage: ~ 10 TB object storage (dense checkpoint corpus: ~ 150 checkpoints/run $\times$ ~ 150 runs, plus probe trajectories), retained 12 months and released publicly; ~ 2 TB shared filesystem for active working sets.

Datasets: public only — FineWeb-Edu / The Pile subsets for pretraining, plus programmatically ablated variants we construct (the manufactured negatives). No private or licensed data.

Preemptible-friendly by construction: our runners are already idempotent with checkpoint-resume (built for a home GPU that gets interrupted). We plan $\approx 70\%$ of fleet training on preemptible instances, with on-demand reserved for latency-sensitive probing and the blind-gate runs.

3. Expected costs (Nebius pricing)

Item	Rate (Nebius)	Cost
$\sim 6,800$ H100-h preemptible (70%)	\$2.15 / GPU-h	\$14,620
$\sim 2,900$ H100-h on-demand (30%)	\$3.85 / GPU-h	\$11,165
Object storage, 10 TB $\times$ 12 mo	\$0.0147 / GiB-mo	\$1,805
Shared filesystem, 2 TB $\times$ 12 mo	\$0.0800 / GiB-mo	\$1,966
CPU/eval nodes, analysis + harness CI	—	$\sim \$1,500$
Core program subtotal		$\sim \\$31,100$
Extended capability tracks, 1B fleet expansion, public-corpus hosting through release		$\sim \$18,900$
Total requested		\$50,000

Scoping flexibility: a \$15,000 award delivers the single-capability core (induction track, 70M–410M fleets with negatives, public corpus). \$30,000 adds the 1B validation fleet and the gap-origin study. \$50,000 delivers all three capability tracks and the full public benchmark. We will report actual spend against these lines.

4. Project timeline with key milestones

Phase	Months	Milestone (deliverable)
M1	1–2	Infrastructure port: probe harness + idempotent multi-seed runners on Nebius; pilot 70M fleet (5 seeds) reproducing the induction phase change on real text. <i>Gate: phase change resolvable at our checkpoint cadence.</i>
M2	2–4	Full 70M–160M fleets (30 seeds/config) + manufactured negatives; scored benchmark v1: lead @ FA $\leq 5\%$, conformal coverage. <i>Pre-registered before launch.</i>
M3	4–6	410M fleets; blind gate #1 on a never-seen configuration. <i>Frozen rule, commit-stamped, one scoring pass.</i>
M4	6–8	Capability track 2 (in-context learning battery): precursor identification, fleets, negatives via targeted ablation.
M5	8–10	1B validation fleet + blind gate #2; gap-origin scaling study (is the precursor→emergence gap set by optimization time, tokens, or competition for gradient?).
M6	10–12	Capability track 3 (arithmetic/syntactic); public corpus + harness release; flagship paper submission.

Every milestone carries pre-registered predictions and kill criteria committed to a public repository *before* the corresponding runs execute. This is our standing practice, not an aspiration: our current program has a public commit chain in which every freeze precedes its data, and three pre-registered kill criteria have fired and been reported.

5. Anticipated outcomes

- (a) **The first multi-seed capability-emergence corpus:** ~ 150 LM pretraining runs (70M–1B) with dense checkpoints, circuit-probe trajectories, and — uniquely — manufactured capability-blocked negatives enabling honest false-alarm accounting. Released publicly.
- (b) **A scored forecasting benchmark + harness:** event definitions, lead-time at controlled FA, conformal interval scoring, and baseline implementations (loss rules, config equation). Reusable by any lab for any capability with a known precursor.
- (c) **The scaling answer:** does per-seed forecastability survive at 1B parameters on real text, and does the precursor→emergence gap remain approximately configuration-invariant (as it did across our first blind shift: $975 \rightarrow 1,012$ steps)? Either answer is decision-relevant for monitoring.
- (d) **A deployable recipe** for labs: identify the precursor circuit; log it densely; manufacture blocked negatives by ablating the capability’s training signal; calibrate an anchored conformal interval on a seed fleet; freeze; monitor; re-certify after recipe changes.
- (e) **Honest boundaries**, reported either way: our discipline has killed three of our own promising results (a label-free probe that failed confirmation; cross-regime calibration transfer; a domain where no forecaster met the FA cap). Negative outcomes will be published with the same rigor as positive ones.

6. Potential publication plans

The flagship paper (“*Capability Emergence Can Be Forecast: Per-Seed, In Advance, With Calibrated Intervals, Certified False Alarms, and a Blind Pre-Registered Gate*”) is written and internally

complete at small scale; the Nebius-funded scale-up becomes its principal results section, targeting **ICLR/NeurIPS** with an arXiv preprint at submission. The corpus and harness ship as a **public benchmark release** (dataset card + leaderboard-style harness). Two follow-ons are natural: a methods paper on manufactured negatives for capability monitoring, and a mechanistic paper on the gap-origin result. Prior output from this line: three arXiv papers on the underlying representational-timing mechanism (one endorsed by the author of the weight-norm delay law we build on), all with the same regeneration-and-pre-registration discipline.

7. Collaboration opportunities with Nebius

- **Co-authorship / acknowledgment:** Nebius credited on the benchmark release and papers; open to co-authorship with Nebius researchers contributing to design or analysis.
- **Hosting the public corpus** on Nebius object storage — a visible, citable artifact (~ 10 TB, permanently referenced by every paper that uses the benchmark) demonstrating Nebius infrastructure in open science.
- **Preemptible-workload case study:** our runners are idempotent and checkpoint-resumable by design. We are glad to publish a technical note with Nebius on running large multi-seed scientific fleets economically on preemptible capacity — directly useful to other Nebius research users.
- **Talks/blog:** happy to present results at Nebius research events or write a guest technical post; the “can we see capabilities coming?” framing has broad interest.
- **Early access for Nebius customers:** the forecasting harness is infrastructure-agnostic but ships with Nebius-ready launch configs.

Why this applicant: an independent researcher with a complete, reproducible research program built on one RTX 3080 — five projects, three arXiv papers, a public commit chain where every prediction is timestamped before its data, and a headline result ($\rho = 0.977$ per-seed forecasting, 10/10 blind gate) already in hand. The compute is the only remaining bottleneck between this and a benchmark the field can use.